

Next-generation Bayesian local robustness

Ryan Giordano (rgiordano@berkeley.edu, UC Berkeley)

Berkeley Neyman Seminar

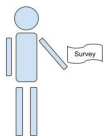
September 2026

The basic problem

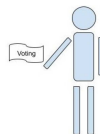
We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.



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(This is the “Data”, $\mathcal{D}$)



Observe x_j for $j = 1, \dots, N_T$

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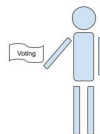
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The problem is that the populations may be very different, maybe leading to bias.



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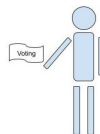
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This talk will study “Multilevel regression and poststratification” estimators:

- Set $\mathbb{E}[y|x, \theta] = m(\theta^\top x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\mathcal{D})$ with Markov Chain Monte Carlo (MCMC)
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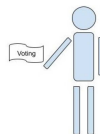
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Problem: MCMC is time consuming and the regression model is surely misspecified.

Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

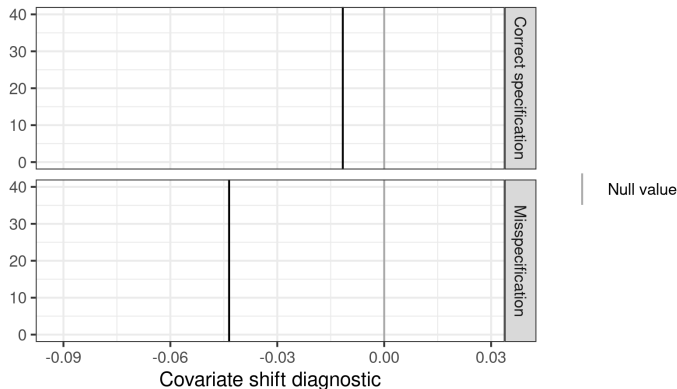


Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecified model is more extreme.

Problem: What is “large enough” to be real evidence of misspecification?

Possible answer: Frequentist confidence intervals via the parametric bootstrap.

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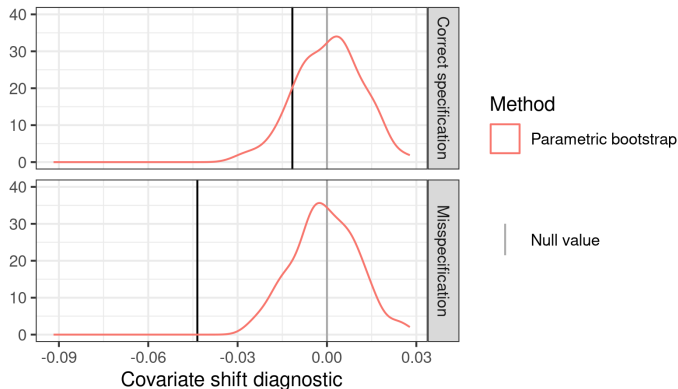


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The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.

Possible answer: Infinitesimal jackknife confidence intervals? [Giordano and Broderick, 2026]

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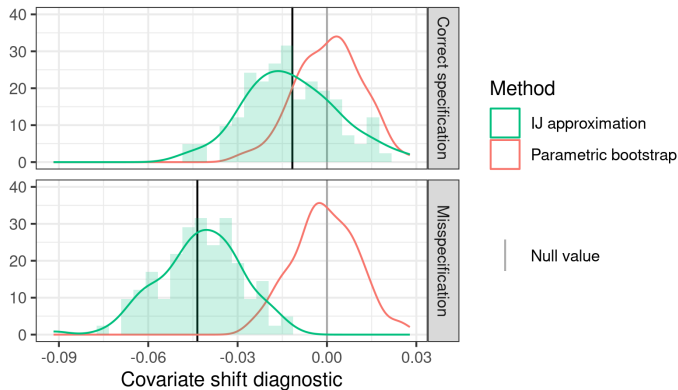


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IJ confidence intervals are fast and often work well for posterior means.
But they are not accurate for this more complex diagnostic (high Monte Carlo error).

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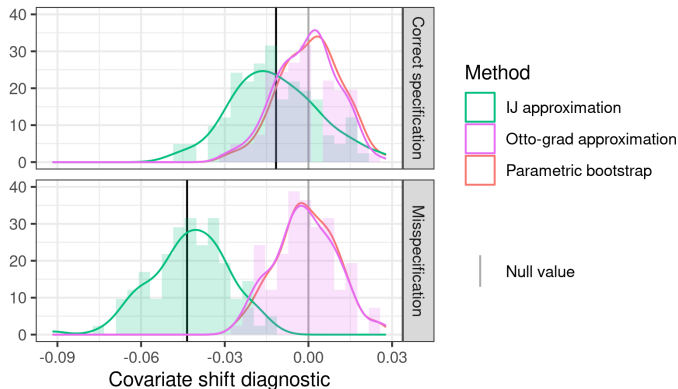


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Answer: Our flow-based local sensitivity measures (“Otto grad”) produce good approximations to the bootstrap at negligible computational expense.

- Bayesian local regression specification checks
 - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
 - (With Tamara Broderick)
- Flow-based Bayesian local robustness: “Otto grad”
 - (With Lucas Schwengber)

TODO: put in pictures

Bayesian local regression specification checks

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j) \right]}_{\text{Want to check specification without re-running MCMC}}.$$

Some existing local approaches:

- Specify an alternative model + test with Bayes factor [Cabral et al.]
 - Bayes factors can depend strongly on priors (Lindley's paradox)
 - Articulating a family of models is inconvenient
 - Doesn't take the target into account
- Use out-of-sample checking [Vehtari et al., Kuh et al.]
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The model is definitely misspecified, we want to check whether it's *importantly* misspecified.

Local specification checks

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Let $\mathcal{P}(x, y)$ denote a generic joint density of x, y .

Let $\hat{\mathcal{P}}_S$ the empirical distribution on the survey data.

1. Write our estimand as a functional $\mu(\mathcal{P})$ that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x) \mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x)) \mathcal{P}(y|x)$
4. Compute the derivative $\Delta(w, \mathcal{P}) = \partial \mu(\mathcal{P}_\epsilon) / \partial \epsilon$
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Lots of similar ideas in causal inference, kernel learning, transfer learning [Bühlmann, Rojas-Carulla et al., Muandet et al.]. This is just a local Bayesian diagnostics version.

Bayesian sensitivity analysis

Consider a family of distributions $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$ and a test function $\phi(\theta)$. Then, by interchange of integration and differentiation,

$$\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) .$$

Let the regression log likelihood be $\ell(y|x, \theta)$, so that

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp \left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta) \right) \text{ and } \mathcal{P}^\epsilon(\theta|\text{Data}) \propto \mathcal{P}(\theta|\mathcal{D}) \exp \left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta) \right)$$

- $\mathcal{P}(\theta) = \mathcal{P}(\theta|\mathcal{D})$
- $\phi(\theta) = \frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j)$ (so that $\hat{\mu} = \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)]$)
- $h(\theta) = \sum_{i=1}^{N_S} w(x_i) \ell(y_i|x_i, \theta) \Rightarrow$

Our diagnostic is then given by

$$\Delta(w, \hat{\mathcal{P}}_S) := \left. \frac{\partial \hat{\mu}}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta|\mathcal{D})} \left(\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j), \sum_{i=1}^{N_S} w(x_i) \ell(y_i|x_i, \theta) \right)$$

We expect $\Delta(w, \hat{\mathcal{P}}_S)$ to be close to zero when the model is correctly specified and posterior estimation is accurate.

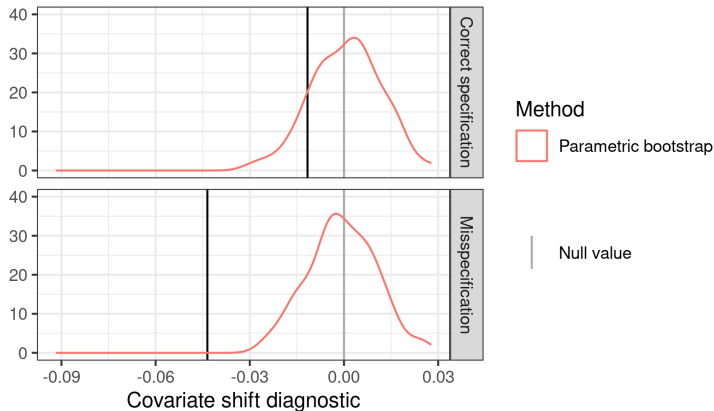


Figure 2: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

Model 0 (correct) : $y \sim \text{Logistic}(1 + X_1 + X_2 + X_3 + X_4 + X_5)$

Model 1 (misspecified) : $y \sim \text{Logistic}(1 + X_3 + X_4 + X_5)$

Compute diagnostic with $w(x) = \mathbb{I}(X_2 > 0.878)$

Frequentist variance with the infinitesimal
jackknife

Why frequentist variance?

For the remainder of the talk, focus on approximating the *frequentist* sampling distribution under $x_i, y_i \stackrel{iid}{\sim} \mathcal{P}_S(x, y)$ of quantities like our diagnostic:

$$\Delta = \text{Cov}_{\mathcal{P}(\theta|\mathcal{D})} \left(\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j), \sum_{i=1}^{N_S} w(x_i) \ell(y_i | x_i, \theta) \right)$$

No need to commit to our covariate shift diagnostic to care!

- Frequentist variability of *any* model checking diagnostic is interesting
- Frequentist variability remains meaningful under model misspecification (Huggins and Miller, IJ papers)

We will form *local approximations* computable only from *the original posterior*.

We will compare with computationally expensive bootstrap:

- Compute Δ on the original dataset
- For $B = 1, \dots, 100$:
 1. Sample $(x_1^*, y_1^*), \dots, (x_{N_S}^*, y_{N_S}^*)$ (parametric or nonparametric bootstrap)
 2. Run MCMC to estimate $\mathcal{P}(\theta | (x_1^*, y_1^*), \dots, (x_{N_S}^*, y_{N_S}^*))$
 3. Use the draws to estimate Δ_B^*
- Compare Δ with quantiles of $\Delta_B^* - \Delta$

Data re-weighting.

Augment the problem with *data weights* $w_1, \dots, w_N$. We can write $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$.

$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta) \qquad \log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

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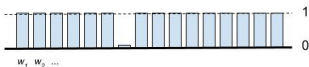
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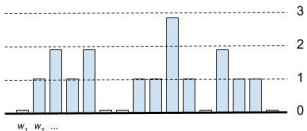
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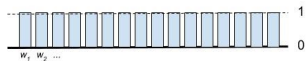
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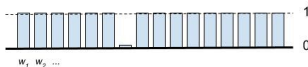
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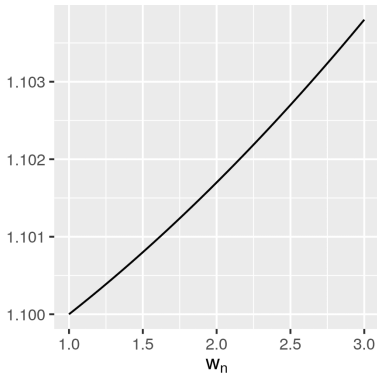
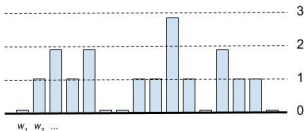
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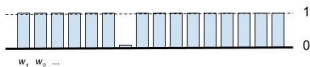
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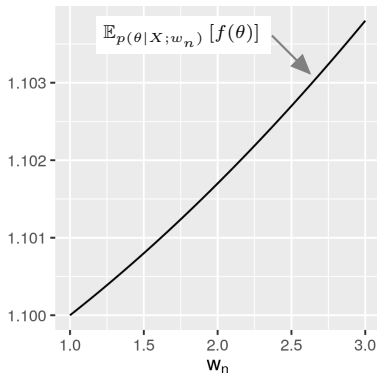
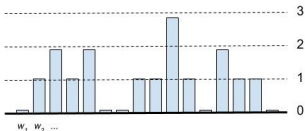
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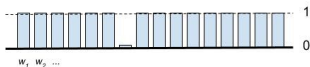
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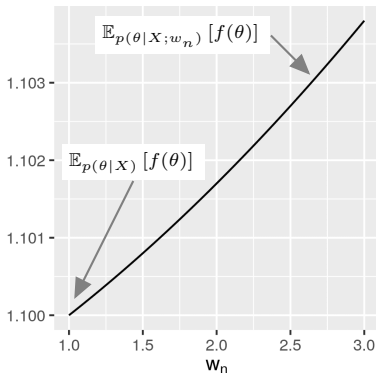
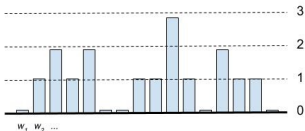
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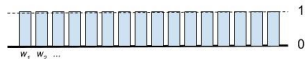
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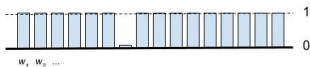
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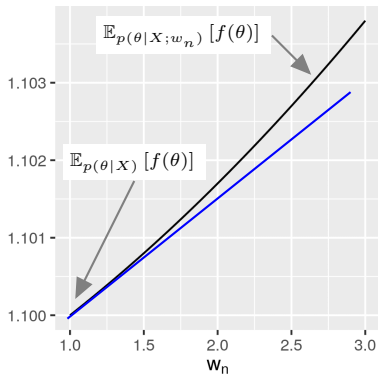
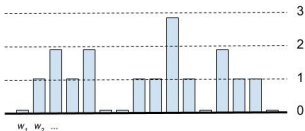
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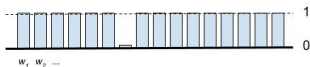
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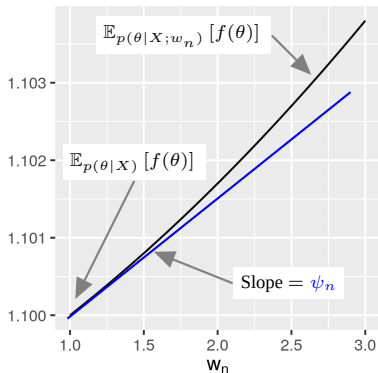
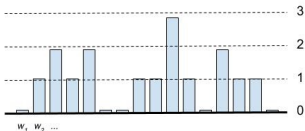
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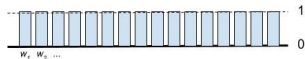
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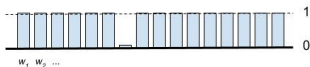
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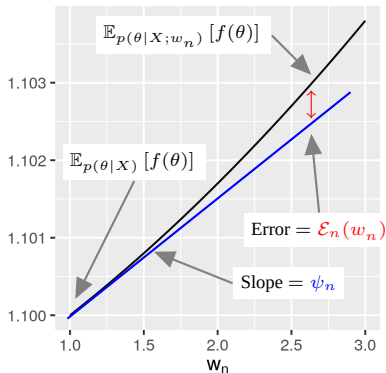
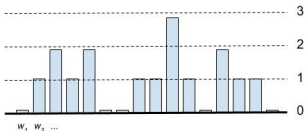
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Leave-one-out weights:



Bootstrap weights:



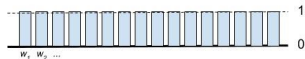
Data re-weighting.

Augment the problem with *data weights* $w_1, \dots, w_N$. We can write $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$.

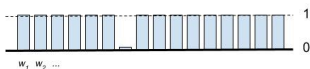
$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta)$$

$$\log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

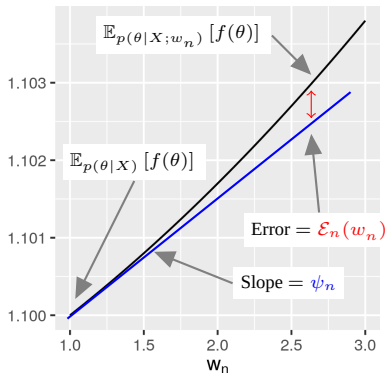
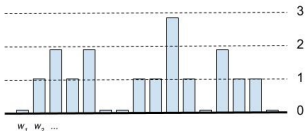
Original weights:



Leave-one-out weights:



Bootstrap weights:



The re-scaled slope $N\psi_n$ is known as the “influence function” at data point x_n .

$$\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)] - \mathbb{E}_{\mathcal{P}(\theta|X)} [f(\theta)] = \sum_{n=1}^N \psi_n (w_n - 1) + \mathcal{E}_n(w)$$

How can we use the approximation?

Assume the **slope** is computable and **error** is small.

$$\mathbb{E}_{p(\theta|X;w)} [f(\theta)] - \mathbb{E}_{p(\theta|X)} [f(\theta)] = \sum_{n=1}^N \psi_n(w_n - 1) + \mathcal{E}_n(w)$$

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Bootstrap. Draw bootstrap weights $w \sim \mathcal{P}(w) = \text{Multinomial}(N, N^{-1})$.

$$\text{Bootstrap variance} = \text{Var}_{\mathcal{P}(w)} \left(\mathbb{E}_{p(\theta|X;w)} [f(\theta)] \right)$$

How can we use the approximation?

Assume the **slope** is computable and **error** is small.

$$\mathbb{E}_{p(\theta|X;w)} [f(\theta)] - \mathbb{E}_{p(\theta|X)} [f(\theta)] = \sum_{n=1}^N \psi_n(w_n - 1) + \mathcal{E}_n(w)$$

Bootstrap. Draw bootstrap weights $w \sim \mathcal{P}(w) = \text{Multinomial}(N, N^{-1})$.

$$\begin{aligned} \text{Bootstrap variance} &= \text{Var}_{\mathcal{P}(w)} \left(\mathbb{E}_{p(\theta|X;w)} [f(\theta)] \right) \\ &= \text{Var}_{\mathcal{P}(w)} \left(\sum_{n=1}^N \psi_n(w_n - 1) + \mathcal{E}_n(w) \right) \\ &= \frac{1}{N^2} \sum_{n=1}^N \left(\psi_n - \bar{\psi} \right)^2 + \text{Term involving } \mathcal{E}_n(w) \text{ for } n = 1, \dots, N \\ &\approx \frac{1}{N} \underbrace{\left(\frac{1}{N} \sum_{n=1}^N \left(\psi_n - \bar{\psi} \right)^2 \right)}_{\text{"Infinitesimal jackknife variance estimate"}} \end{aligned}$$

Bayesian von-Mises Expansion

Define the “generalized posterior” functional [Giordano and Broderick, 2026]:

$$T(\mathcal{P}, N) := \frac{\int g(\theta) \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}{\int \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}.$$

Recall that $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [g(\theta)] = T(\hat{\mathcal{P}}_S, N_S)$

To derive frequentist variance estimates, study the *von Mises expansion*.

Let $\mathcal{P}_t = t\hat{\mathcal{P}}_S + (1 - t)\mathcal{P}_S$. Then form the series expansion in t :

$$\begin{aligned} \sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [g(\theta)] - T(\mathcal{P}_S, N_S) \right) &= \sqrt{N_S} \left(T(\hat{\mathcal{P}}_S, N_S) - T(\mathcal{P}_S, N_S) \right) \\ &= \sqrt{N_S} \left. \frac{\partial T(\mathcal{P}_t, N_S)}{\partial t} \right|_{t=0} (1 - 0) + \mathcal{E}(\tilde{t}) \\ &= \underbrace{\sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}). \end{aligned}$$

where

$$\psi_i := N_S \text{Cov}_{\mathcal{P}(\theta|\mathcal{D})} (g(\theta), \ell(y_i|x_i)) \quad \text{and} \quad \tilde{t} \in [0, 1].$$

Bayesian von-Mises Expansion Results

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [g(\theta)] - T(\mathcal{P}_S, N_S) \right) = \sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

- Assume (sketch): A well-behaved MAP *maximum a posteriori* estimator $\hat{\theta}$ exists.
 - The dimension of θ is fixed as $N \rightarrow \infty$
 - The expected log likelihood has a strict maximum at θ_0
 - The observed log likelihood satisfies $\hat{\theta} \rightarrow \theta_0$
 - The expected log likelihood Hessian is negative definite at θ_0
- Assume (sketch): We can apply standard asymptotics.
 - The log prior and log likelihood are four times continuously differentiable
 - The prior is proper, and a technical set of prior expectations are finite
 - The log likelihood and its derivatives are dominated by a square-integrable envelope function for all θ in a neighborhood of θ_∞ .

Theorem 3 [Giordano and Broderick, 2026] (sketch):

$$\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{P} 0 \quad \text{in the Bayesian von-Mises expansion.}$$

Corollary: Let $\frac{1}{N_S} \sum_{i=1}^{N_S} (\psi_i - \bar{\psi})^2 \xrightarrow{P} V$. By the CLT applied to ψ_i ,

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [g(\theta)] - T(\mathcal{P}_S, N_S) \right) \xrightarrow{d} \mathcal{N}(0, V).$$

Recall that our covariate shift diagnostic is given by

$$\Delta(w, \hat{\mathcal{P}}_S) = \text{Cov}_{\mathcal{P}(\theta|\mathcal{D})} \left(\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j), \sum_{i=1}^{N_S} w(x_i) \ell(y_i | x_i, \theta) \right) = \sum_{i=1}^{N_S} \psi_i w(x_i).$$

In other words, we are checking whether the influence function aligns with any of the regressors.

To compute the parametric bootstrap, it's a simple change. (Cite MrP as a special case.)

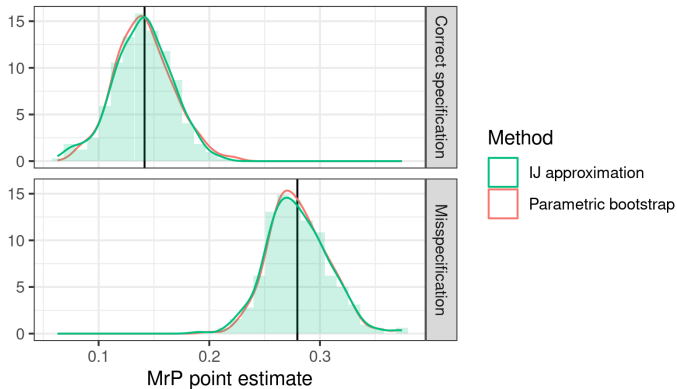


Figure 3: MrP Posterior Mean

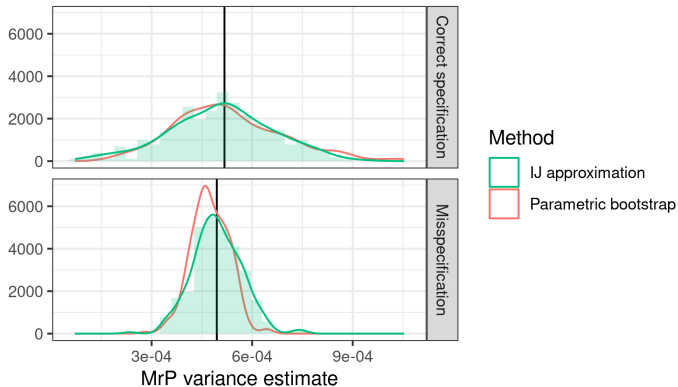


Figure 3: MrP Posterior Variance

Data re-weighting.

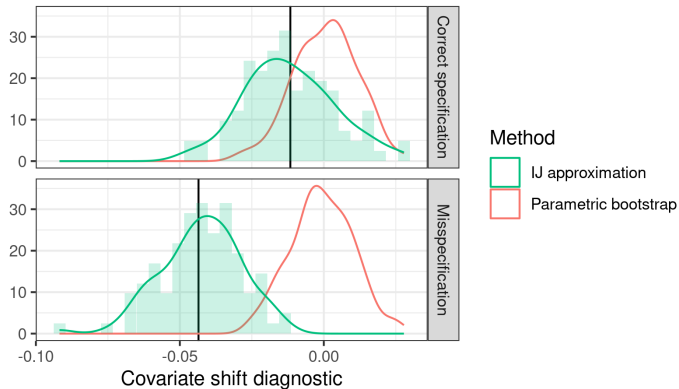


Figure 3: Covariate Shift Diagnostic

Need something better for noisier / more complex posterior quantities!

Local robustness with flows

Why flows for sensitivity analysis?

The IJ is built around linearizing posterior expectations:

$$\mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [g(\theta)] \approx \mathbb{E}_{\mathcal{P}} [g(\theta)] + \left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [g(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} (\epsilon - 0),$$

where the derivative is estimated — with noise — from MCMC draws.

This has some shortcomings:

- Potential MCMC variance, not obvious how to smooth
- Linear extrapolation can lead to impossible values (e.g. negative variances)
- Complex quantities like posterior quantiles are hard to linearize
- Bayesians like to work with draws, not analytic expectations (CITE)

Idea: Instead of linearizing expectations, learn a flow from $\mathcal{P}$ to $\mathcal{P}^\epsilon$.

There's a large literature on learning flows between distant distributions. (CITE)

Our problem is easier:

- Only need “local” flows
- Our perturbations have a lot of structure (e.g. bootstraps)

Consider a family of distributions $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$ and a test function $\phi(\theta)$. Then, by interchange of integration and differentiation,

$$\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) .$$

We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ , $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

Suppose there is a scalar-valued potential function $v(\theta)$ such that

$$\begin{aligned} f^\epsilon(\theta) &= \theta + \epsilon \nabla v(\theta) + O(\epsilon^2) \quad \text{as } \epsilon \rightarrow 0. \text{ Then:} \\ \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] &= \mathbb{E}_{\mathcal{P}(\theta)} [\phi(f^\epsilon(\theta))] \Rightarrow \\ \left. \frac{\partial}{\partial \epsilon} \right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] &= \left. \frac{\partial}{\partial \epsilon} \right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}(\theta)} [\phi(\theta + \epsilon \nabla v(\theta))] \\ \parallel &\parallel \\ \mathbb{E}_{\mathcal{P}(\theta)} [\phi(\theta) h(\theta)] &= \mathbb{E}_{\mathcal{P}(\theta)} [\nabla \phi(\theta)^\top \nabla v(\theta)] \end{aligned}$$

- When does such a $v(\theta)$ exist?
- How can we find it in practice?

When does such a $v(\theta)$ exist?

Consider a family of distributions $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$ and a test function $\phi(\theta)$. Then, by interchange of integration and differentiation,

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We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ , $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

When does such a $v(\theta)$ exist?

Theorem (Schewengber 2026): Assume that

- $\log \mathcal{P}(\theta) \in C^2(\mathcal{R}^d)$
- $\lim_{\theta \rightarrow \infty} \left(\frac{1}{2} \|\nabla \log \mathcal{P}(\theta)\|^2 - \Delta \log \mathcal{P}(\theta) \right) = +\infty$.
- $\mathbb{E}_{\mathcal{P}(\theta)} [h(\theta)^2] < \infty$

Then there exists a solution v to the differential equation

$$-\mathcal{L}_{\mathcal{P}}(v) := \nabla v^\top \nabla \log \mathcal{P} + \Delta v = h$$

such that $\|\nabla v\|_{L^2(\mathcal{P})} < \infty$, and the flow $\theta_\epsilon = \theta + \epsilon \nabla v(\theta)$ satisfies $\theta_\epsilon \sim \mathcal{P}^\epsilon(\theta)$ for small ϵ .

Note that $\mathcal{L}_{\mathcal{P}}$ is the drift operator, generator of Langevin diffusion, trace of the Stein operator.

How can we find $v(\theta)$ in practice?

Fix $h(\theta)$. Specify

- A family of real-valued test functions $\phi_n(\theta)$ for $n \in [N]$
- A family of real-valued basis functions $b_m(\theta)$ for $m \in [M]$

Let $v(\theta; \beta) := \sum_{m=1}^M \beta_m b_m(\theta)$. Then

$$\underbrace{\mathbb{E}_{\mathcal{P}(\theta)} [\phi_n(\theta) h(\theta)]}_{\text{Estimate with MCMC!}} = \sum_{m=1}^M \beta_m \underbrace{\mathbb{E}_{\mathcal{P}(\theta)} [\nabla \phi_n(\theta)^\top \nabla b_m(\theta)]}_{\text{Estimate with MCMC!}}$$

This gives N linear equations in M unknowns. Known as “Otto’s calculus” in differential equations. Formally similar to numerical PDE solutions known as Galerkin methods, except we have a stochastic grid ($\theta \sim \mathcal{P}(\theta)$) rather than a deterministic one.

Unlike standard linear regression:

- We are free to choose ϕ_n and b_m
- Both F and B are measured with noise
- We have a lot of information about the covariance of F and B

Theorem (Schewengber 2026, sketch): The solution gives the best L2 approximation to $\|\nabla v(\theta)\|_{L^2(\mathcal{P})}$ in the span of $\phi_n(\theta)$. When $\mathcal{P}$ is Gaussian, the ordered Hermite polynomials give the best finite-rank approximation.

Suppose we believe that $\mathcal{P}(\theta|\mathcal{D})$ is approximately Gaussian.

Take the basis functions $b_m(\theta) = \theta_m$ (1st Hermite polynomial) and test functions $\phi_n(\theta) = b_m(\theta)$. Then the optimal flow is given by

$$\theta_\epsilon = \theta + \epsilon \underbrace{\text{Cov}_{\mathcal{P}(\theta|\mathcal{D})}(\theta, h(\theta))}_{\left. \frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[\theta]}{\partial \epsilon} \right|_{\epsilon=0}}$$

This just translates the posterior in the direction of the mean changes.

Suppose we have $\theta_m \sim \mathcal{P}(\theta|\mathcal{D})$, and bootstrap weights w_i . Set

$$\theta_m^* = \theta_m + \frac{1}{N_S} \sum_{i=1}^{N_S} w_i \psi_i$$

and compute any posterior quantities from these samples.

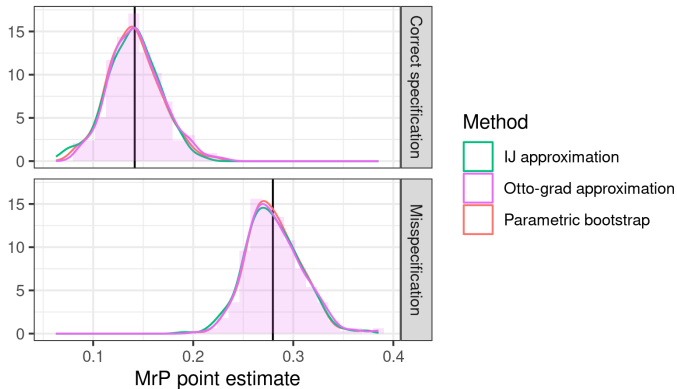


Figure 4: MrP Posterior Mean

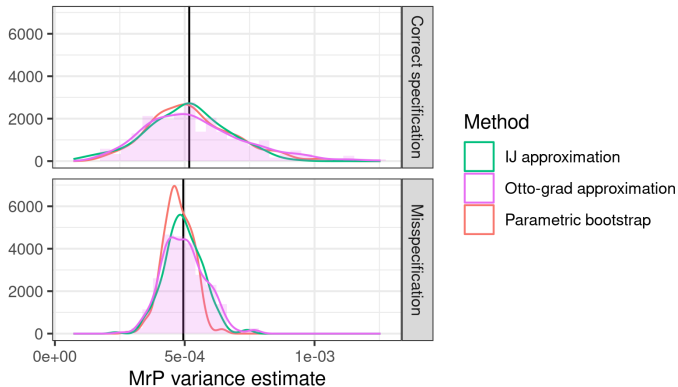


Figure 4: MrP Posterior Variance

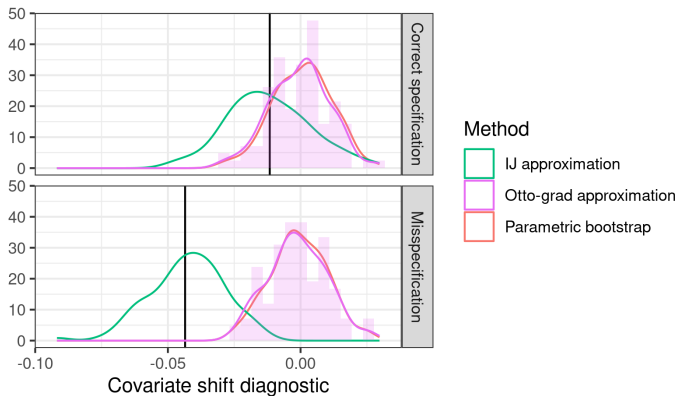


Figure 4: Covariate Shift Diagnostic

Understanding the gap between IJ and Otto is ongoing work!

- Bayesian local regression specification checks
 - Use invariance to covariate shift as a diagnostic for Bayesian hierarchical models
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
 - Use sensitivity of mean parameters to speed up bootstrapping MCMC
- Flow-based Bayesian local robustness: “Otto grad”
 - Use easy-to-compute approximate flows to speed up bootstrapping MCMC

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